

STEP 13 — COMPOSITE INPUTS AT FULL POPULATION

HDI-N project | Remaining digital and HDI inputs, and the operator comparison | Follows Step 12 (reissued)

OBJECTIVE

Step 12 established the facade quantities for 184 economies. This step retrieves the remaining inputs needed to rebuild the HDI-N composite at the same scale, and settles one open methodological question: whether the country ordering still depends on how many components of the neutrosophic triple enter the aggregation. Task 2 carries a hard gate, because the coverage of the high-technology export series determines whether the composite is viable at full population at all. Retrieval and computation only — no prose and no paper edits are authorised under this memo.

TASKS

Task	What to do	Deliverable	Gate / verification
1	Retrieve fixed-broadband subscriptions per 100 inhabitants (ITU DataHub series i992b) for all reporting economies, reference year 2023. Where 2023 is absent, take the most recent value within 2020–2023 and record the year used. Outside that window, record as missing.	Step13_ITU_Broadband_Full.csv Columns, in this order: ISO3, Country_ITU, Broadband_per100, Year_Used, Status, Retrieval_Date.	Confirm in writing that the series pulled is i992b (subscriptions per 100 inhabitants) and NOT i4213tfbb, which carries the same unit label but reports raw subscription counts. Submit the DataHub series page with the retrieval date.
2	Retrieve high-technology exports as a percentage of manufactured exports (World Bank WDI, series TX.VAL.TECH.MF.ZS) for all economies, reference year 2023, same window rule as Task 1. Before computing anything else, report the coverage count.	Step13_WDI_HiTech_Full.csv Columns, in this order: ISO3, Country_WDI, HiTech_Pct, Year_Used, Status, Retrieval_Date. Plus a one-line coverage statement: the number of economies with any value in 2020–2023.	GATE. Send the coverage count on its own, before starting Tasks 4 and 5, and wait for acknowledgement. If coverage is materially below 130 economies the design of the composite changes and the remaining tasks may be revised. Also report the ten economies with the highest HiTech_Pct, with values.
3	Assemble two country-attribute joins: (a) the standard HDI dimensional indices — health, education and income — for all economies from the UNDP Human Development Report 2023–24, on the UNDP goalpost scale; (b) the UN M49 region and sub-region, plus the World Bank region, for all economies.	Step13_HDR_Dimensions_Full.csv Columns: ISO3, Country_HDR, I_health, I_education, I_income, HDI_Standard, HDI_Rank_Standard, Retrieval_Date. Step13_Region_Lookup.csv Columns: ISO3, Country, M49_Region, M49_Subregion, WB_Region.	State the HDR table and edition used. Report the number of economies with all three dimensional indices present.
4	Merge Tasks 1–3 onto the Step 12 facade file on ISO3. Define the computable set: economies with EGD, penetration, broadband, high-technology exports and all three HDI dimensions present and strictly positive. Normalise: $I_{pen} = U / 100$; $I_{bb} = B / \max(B)$; $I_{ht} = H / \max(H)$, both maxima taken over the computable set and over no other set.	Step13_Master_Inputs.csv One row per economy in the computable set, carrying: ISO3, Country, region columns, all raw inputs with their Year_Used columns, I_{pen} , I_{bb} , I_{ht} , EGD, OSI, TII, p, F_egov, I_egov, and the three HDI dimensional	Report the size of the computable set and a full list of economies excluded, each with the single input that caused the exclusion. Economies reporting exactly zero on any input are excluded by the same rule — list them separately rather than

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		indices. Plus a stated normalisation base: the value of max(B) and max(H) and which economy sets each.	substituting a small constant.
5	Compute the e-government score under three operators, using the same normalisation base throughout: $A \ T_{adj} = EGD \times p$ $B \ T_{adj} = EGD \times p \times (1 - I_{egov})$ $C \ s = (2 + EGD - I_{egov} - F_{egov}) / 3$ For each, substitute at the same point and compute $I_{digital}$ as the fourth root of $(I_{pen} \times I_{bb} \times T_{adj} \times I_{ht})$, then HDI-N as the fourth root of $(I_{health} \times I_{education} \times I_{income} \times I_{digital})$. Nothing else changes between the three.	Step13_HDIN_Operators.csv Columns: ISO3, Country, $I_{digital_A}$, HDIN_A, Rank_A, $I_{digital_B}$, HDIN_B, Rank_B, $I_{digital_C}$, HDIN_C, Rank_C, HDI_Standard, Rank_Standard. Sorted descending by HDIN_A.	Do not reconcile the three orderings or select among them. Where they disagree, the disagreement is the result.
6	Reconciliation and comparison report. Figures only, no interpretation.	Step13_Operator_Report.md Covering: (a) Spearman correlation for each of the three operator pairs, and of each against the standard HDI; (b) the count of economies moving 3 or more, and 5 or more, ranks between A and B, and between A and C, with those economies named; (c) the twenty original panel economies with their rank under A, B and C. Plus: a re-retrieval of Bangladesh's EGD 2024 from its UN DESA country page, reported to the precision published there.	The Bangladesh figure resolves a discrepancy between the project panel file (0.656) and the Step 12 retrieval (0.6570). Report what the source page shows. Do not adjust either existing file.

WATCH-OUTS

- The three operators are alternative specifications, not a draft and two checks. Compute all three to the same precision, from the same inputs, and report all three in full.
- No imputation, anywhere. The geometric mean is undefined if any input is missing and collapses to zero if any input is zero, so an economy short one input leaves the computable set. It does not receive a substituted value, a regional average, or a small positive constant.
- Normalisation maxima are taken over the computable set only, and the same base is used for all three operator variants. Recomputing the maximum separately per variant would make the three orderings incomparable, which is the one thing this step exists to compare.
- Merge on ISO3 only. ITU, WDI, UNDP and UN DESA all use different country-name conventions.
- Task 2's ten-highest list matters: a high share of manufactured exports can reflect assembly and re-export rather than domestic capability, and small economies with a narrow manufacturing base can post extreme values on a small denominator. Report the values; do not exclude or adjust them.
- Scope stops at Task 6. Do not retrieve SWIID, shadow-economy or Gini data, and do not compute disparity scores. That analysis is under separate consideration.
- Column names, column order and sort order are as specified and are not to be varied.

SEQUENCING

Tasks 1, 2 and 3 are independent and can be run in parallel by up to three students. Task 2's coverage count is a hard gate: it is sent on its own and acknowledged before Tasks 4 and 5 begin, because a low count changes what the composite can be. Task 4 requires all of 1, 2 and 3 plus the Step 12 facade file. Task 5 follows Task 4. Task 6 follows Task 5, except for the Bangladesh EGDI re-retrieval, which is independent and can be done at any point.

Return the six files together as a single submission, with the exception of the Task 2 coverage count, which comes early and on its own.